

Pseudocode: C2 Communication Blueprint

1. Global Setup & Security Rules

- Define the Server IP address and listening port.
- Define a Pre-Shared Encryption Key (AES) for secure communication.
- Define a base wait time (e.g., 60 seconds) and a jitter percentage (e.g., 10%) to randomize client check-ins.

2. The Server (Command Center) Logic

- **Start Server:**
 - Initialize a database to store connected agents, queued commands, and task results.
 - Open a secure (TLS) listening port.
- **Continuous Server Loop:**
 - **Wait** for an incoming connection from a client.
 - **When a connection occurs:**
 - Receive the data and attempt to decrypt it using the Pre-Shared Key.
 - **If** decryption fails (invalid key or tampered data):
 - Immediately drop the connection to prevent unauthorized probing.
 - **If** the decrypted message is a new client checking in:
 - Log the client's unique ID and system info into the database.
 - **If** the decrypted message is a routine check-in:
 - Check the database for any queued commands for this specific client ID.
 - **If** commands exist: Encrypt them and send them to the client.

- **If** no commands exist: Encrypt a *Sleep* command and send it.
- **If** the decrypted message contains task results:
 - Save the output or error messages to the database for the operator to view.
- Close the connection safely and wait for the next client.

3. The Client (Agent) Logic

- **Start Client:**
 - Generate a Unique ID based on the machine's hardware.
 - Gather basic system info (OS version, user privileges).
- **Continuous Client Loop:**
 - Calculate the "Sleep Time" (Base Wait Time + or - the Jitter percentage).
 - **Wait/Sleep** for the calculated duration.
 - **Try** to connect to the Server IP and Port:
 - **If** the connection fails (server offline):
 - Silently log the network error locally and loop back to the "Sleep" phase. Do not crash.
 - **If** the connection succeeds:
 - Encrypt the Unique ID and send it to the server to check in.
 - Receive the encrypted response and decrypt it.
 - **If** the instructions say "Sleep":
 - Close the connection and loop back to the "Sleep" phase.
 - **If** the instructions contain system commands:
 - **Try** to execute the commands locally one by one.
 - **If** a command succeeds: Capture the output.
 - **If** a command fails (e.g., access denied): Capture the error message.

- Encrypt the captured output/errors and send them back to the server.
- Close the network connection and restart the loop.